

STAT 709: Homework 1

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Due: September 20, 2026, 11:59 p.m.

1. Jun Shao's book: Chapter 1, Exercises 7, 8, and 10.

2. For a matrix $A \in \mathbb{R}^{n \times m}$, recall that the operator norm is the largest singular value: $\|A\|_{\text{op}} = s_{\max}(A)$. Another useful norm is the **Frobenius norm**, defined by

$$\|A\|_{\text{F}} = \left(\sum_{i=1}^n \sum_{j=1}^m A_{ij}^2 \right)^{1/2}.$$

Let $r = \text{rank}(A)$, and assume $1 \leq r \leq \min\{n, m\}$. Write the nonzero singular values as $s_1(A) \geq \dots \geq s_r(A) > 0$.

- (i) Prove that $\|A\|_{\text{F}}^2 = \sum_{k=1}^r s_k(A)^2$.
- (ii) Prove that $\|A\|_{\text{op}} \leq \|A\|_{\text{F}} \leq \sqrt{r} \|A\|_{\text{op}}$.
- (iii) Show that $\|A\|_{\text{op}} = \|A\|_{\text{F}}$ if and only if $r = 1$. Characterize the nonzero eigenvalues of $AA^{\top}$ and $A^{\top}A$ when $\|A\|_{\text{F}} = \sqrt{r} \|A\|_{\text{op}}$.

3. (Optional)¹ Recall the example from the lecture: let $U \sim \text{Unif}([0, 2\pi])$, and define $X = \cos U$ and $Y = \sin U$. We want to describe their joint distribution $\mu_{X,Y}$ on $\mathbb{R}^2$.

(1) Define $f(u) = (\cos u, \sin u)$ for $u \in [0, 2\pi]$. Prove that f is measurable as a map from $([0, 2\pi], \mathcal{B}([0, 2\pi]))$ to $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. You may cite an appropriate theorem. Use the definition of an induced measure to express $\mu_{X,Y}(B)$ for $B \in \mathcal{B}(\mathbb{R}^2)$.

(2) Alternatively, construct the distribution as a measure on the unit circle. Let

$$S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}, \quad A_{u,v} = \{(\cos z, \sin z) : z \in (u, v)\}, \quad u < v.$$

Let $\mathcal{B}(S)$ be the σ -algebra generated by all sets $A_{u,v}$, where $u, v \in \mathbb{R}$ and $u < v$.

- (i) Prove that there is a unique **probability measure** μ_0 on $(S, \mathcal{B}(S))$ such that

$$\mu_0(A_{u,v}) = \frac{v - u}{2\pi} \quad \text{for every } 0 < u < v < 2\pi.$$

You may cite an appropriate measure extension or uniqueness theorem.

- (ii) For every $B \in \mathcal{B}(\mathbb{R}^2)$, show that $B \cap S \in \mathcal{B}(S)$. Then show that

$$\mu(B) := \mu_0(B \cap S), \quad B \in \mathcal{B}(\mathbb{R}^2),$$

defines a probability measure on $\mathbb{R}^2$.

- (iii) Show that $\mu = \mu_{X,Y}$, where $\mu_{X,Y}$ is the measure from part (1).

¹This exercise is intended for students familiar with measure theory.